2. MATERIALS AND METHODS

This study is based on a comparative genomic analysis between *Homo sapiens* and other mammalian species, using evolutionary distances estimated through phylogenetic methods. The main goal was to investigate evolutionary patterns in orthologous genes, particularly those related to the immune system. The analysis began with a comprehensive list of all human protein-coding genes, from which 1:1 orthologs were identified across 98 mammal species. Each gene was processed individually through an analysis pipeline, described in detail below, which generated evolutionary distance estimates between each species' gene sequence and the corresponding human sequence.

In total, 16,210 genes were analyzed. The analyses were structured in two main approaches: (i) an overall analysis across all genes and species, aiming to identify general divergence patterns. This included identifying genes with the highest and lowest levels of divergence and evaluating divergence across biological pathways; and (ii) targeted analyses focusing on a subset of species and the most divergent genes identified in the overall analysis. This included detecting signals of positive selection at specific sites and mapping divergence patterns within gene interaction networks.

2.1 Evolutionary distance calculation pipeline

An analysis pipeline was developed to process each gene individually, performing sequence collection, alignment, phylogenetic analysis, and distance calculation. The pipeline was implemented using Python and Bash scripts, and the main steps are detailed below.

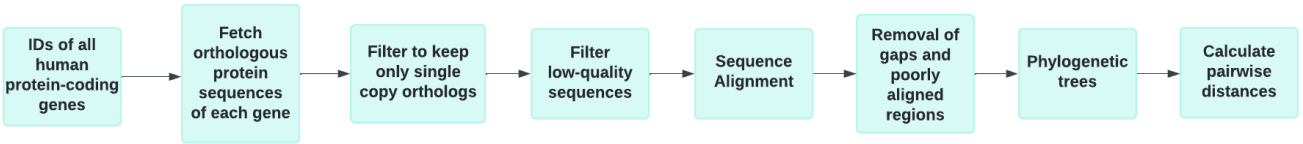

Figure 1: Flowchart from data collection to pairwise distance calculation.

2.1.1 Sequence retrieval

Initially, the BioMart tool from Ensembl (YATES *et al.*, 2016) was used to obtain a list of all human protein-coding genes (release 110, accessed November 2023). From this list, only genes with annotation in the database were selected, ensuring that only those with complete information were considered for subsequent analyses, totaling 19,389 human protein-coding genes.

Using the API of Ensembl, one-to-one orthologs (single-copy orthologous genes) of these human genes in other mammalian species were retrieved from Ensembl Compara (VILELLA *et al.*, 2008). This database provides information on orthology and paralogy relationships among genes from different species. Focusing on single-copy orthologs maintains a direct ancestral relationship between the compared species, minimizing complexities introduced by gene duplications. For each orthologous gene, only the canonical protein sequences were collected. Ensembl defines the canonical sequence as the most biologically relevant transcript, often the longest coding sequence and well-supported by expression data.

All mammalian species available in Ensembl Compara (release 110, accessed November 2023) were considered, totaling 98 species (appendix A - table 3). However, 9 species were removed due to a low number of genes retrieved. Sequences were filtered for quality using a custom script. Sequences containing more than 5% non-standard amino acid characters (e.g., unknown residues or ambiguities) were excluded.

2.1.2 Alignment and Trimming

Each gene's protein sequences were aligned using MAFFT v.7.453 with the *--auto* parameter (KATOH; STANDLEY, 2013). MAFFT is a multiple sequence alignment tool known for its accuracy and speed, which adapts to the size and complexity of the data with the --auto setting. Misaligned regions were trimmed using trimAl v.1.4 with the *automated1* parameter (CAPELLA-GUTIÉRREZ et al., 2009), which automatically removes poorly

aligned or noisy positions from the alignments, enhancing the quality of subsequent phylogenetic analyses. Also, visually poorly aligned or low-quality sequences were excluded.

2.1.3 Calculation of Evolutionary Distances

Evolutionary distances between species pairs were calculated using the Maximum Likelihood (ML) phylogenetic method. ML is a statistical inference method that, in the context of phylogeny, the goal is to find the tree (branch lengths and topology) that maximizes the probability of observing the data given the tree (FELSEINSTEIN, 1981). Therefore, after testing multiple hypotheses, the method selects the tree, together with a substitution model, that would most likely have produced the observed nucleotide or amino acid aligned sequences (POSADA; CRANDALL, 2021).

In this context, the substitution model is applied to describe how residues have changed throughout evolutionary history, defining probabilities that a residue (of amino acid or nucleotide) has changed from one type to another. Protein substitution models, for instance, can incorporate empirical data, assumptions about frequencies, and rates of substitution between sequence residues, helping to model sequence evolution over time more accurately (YANG, 2006; WHELAN; GOLDMAN, 2001; LE; GASCUEL, 2008).

In a ML phylogeny, each branch has a length estimated under a substitution model. In this scenario, the branch length is the expected substitutions per site, and quantifies the amount of change along that branch. To compare the evolutionary distance between two species in a phylogeny, one sums the lengths of the branches along the paths that connect the two species in the tree. This total length is commonly referred to as patristic distance and represents the accumulation of substitutions that separate the two sequences (YANG, 2006; NEI; KUMAR, 2000).

Besides reading patristic distances from an ML tree, pairwise ML distances can be computed directly for each sequence pair under a fitted model. In the Randomized Axelerated Maximum Likelihood (RAxML) software (STAMATAKIS, 2014), the *-f x* option estimates model parameters on a guide tree and then reports ML pairwise distances (branch-length estimates) for all pairs in the alignment.

In this study, pairwise ML distances were used as a comparative measure of divergence between species' sequences. For each gene, genetic distances between pairs of species were calculated using RAxML v.8.2.12 (STAMATAKIS, 2014). Pairwise ML distances were estimated with the *-f x* option together with the *"PROTGAMMAAUTO"* option model, which lets RAxML automatically select the best-fitting amino-acid substitution model for each gene and model among-site rate variation with a discrete gamma distribution.

The methodology involved the following steps:

1. Automatic Model Selection and Phylogenetic Analysis: RAxML tested a range of amino acid substitution models (e.g., DAYHOFF, JTT, WAG, LG), both with and without empirical amino acid frequencies. The PROTGAMMAAUTO parameter enabled the program to choose the model that maximized the likelihood score based on an initial parsimony tree generated internally.

2. Accounting for Rate Heterogeneity: The GAMMA model was used to accommodate rate heterogeneity among sites by incorporating a gamma distribution. This accounts for variations in substitution rates across different positions in the protein sequences, improving the accuracy of distance estimates.

3. Calculation of Pairwise Distances: With the optimal evolutionary model selected for each gene, RAxML computed the ML distances between all pairs of sequences in the alignment for each gene. These distances are expressed as the average estimated number of substitutions per site, providing a continuous measure of genetic divergence between sequences (NEI; KUMAR, 2000)

The result was a set of gene-specific distance matrices, each representing the pairwise evolutionary divergence between species. These distances reflect the average number of substitutions per site between two sequences, as estimated under a specific substitution model that accounts for variation in substitution rates across sites.

After computing pairwise ML distances, species-gene entries with values ≥10 were removed (68 pairs), as such extreme values usually indicate artifacts (e.g., misalignment, incorrect orthology/annotation, severe model saturation). Furthermore, genes lacking orthologous sequences in at least 40 species were removed from the analyses to avoid bias in downstream analyses that summarize each gene by its cross-species median distance. After filters, the final analysis set contained 16,210 genes across 89 species.

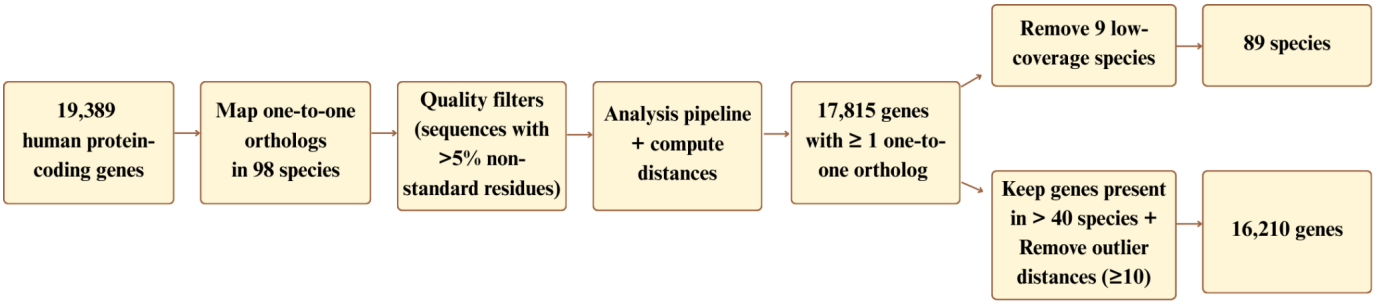

Figure 2: Flow of gene and species filtering. From 19,389 human protein-coding genes, one-to-one orthologs were mapped in 98 mammals. Sequences with >5% non-standard residues were removed and evolutionary distances were computed. After retaining genes with ≥1 ortholog (17,815), removing 9 low-coverage species (final species set: 89), keeping genes present in > 40 species, and excluding 68 distance outliers (distance ≥ 10), the final dataset comprised 16,210 genes.

2.2 Analysis of Evolutionary Distances

Using data from the pipeline described above, a genetic distance database was created for 16,210 single-copy orthologous genes between *Homo sapiens* and 87 other mammalian species. From this database, the following analyses were performed to identify genes with the greatest evolutionary distances and to estimate the distances of immunological pathways based on the average genetic distance of their constituent genes.

The evolutionary distances that were calculated using the phylogenetic method represent the estimated average number of substitutions that occurred between pairs of sequences during evolutionary time. Therefore, a lower distance value (closer to zero) represents a lower estimated average number of substitutions between the sequences. And the higher the value, the greater the number of estimated average substitutions, therefore greater divergence between the sequences.

In this study, the protein sequence of *Homo sapiens* for each gene served as the reference for comparing sequences from other species. Consequently, all reported genetic

distances represent the pairwise evolutionary distance between the ortholog of a given mammal and the human sequence.

2.2.1 Descriptive analysis of gene distances (median-based)

For each gene, the median genetic distance was calculated, as the median is less affected by outliers and provides a more reliable estimate of overall evolutionary divergence. The median is appropriate in evolutionary studies since it offers a picture of typical genetic differences, especially when some species pairs are more divergent due to long evolutionary separation or rapid adaptation.

Genes were then ranked by their median genetic distances and grouped into percentiles. The 15th percentile (most conserved genes) and the 85th percentile (most divergent genes) were identified. This categorization helped highlight genes with distinct evolutionary patterns, allowing for further analysis of their functional significance.

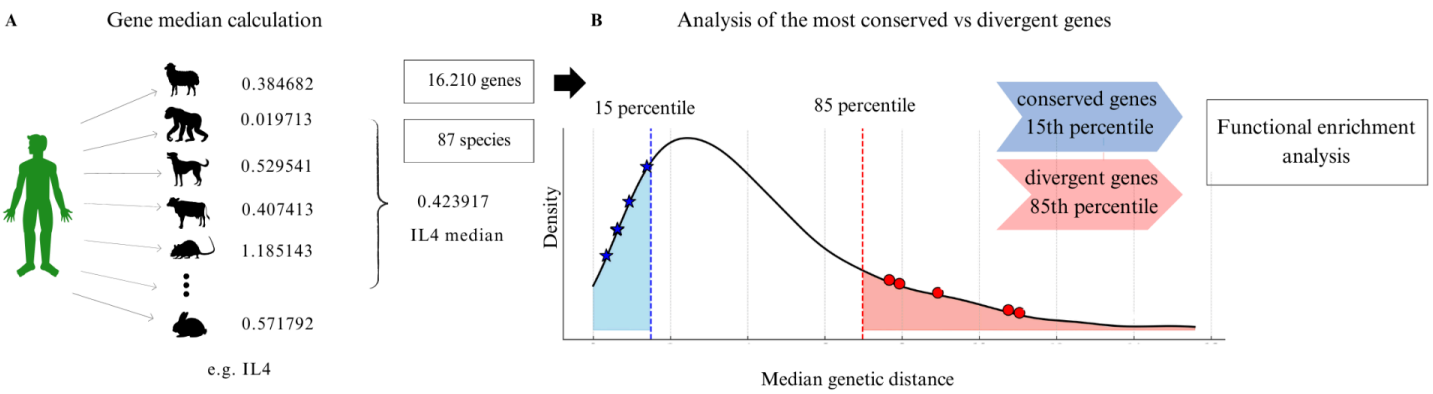

Figure 3: (A) Schematic illustration of how the median genetic distance was calculated for a gene, based on pairwise comparisons between *Homo sapiens* and other mammalian species. (B) Example of the distribution of median genetic distances across the 16,210 genes analyzed. Genes falling within the 15th percentile (most conserved) and the 85th percentile (most divergent) were selected for subsequent functional enrichment analyses.

2.2.2 Functional Enrichment Analysis

To understand the biological functions associated with the most conserved and most divergent genes in the distribution, a functional enrichment analysis was conducted. Using the gene sets from the 15th and 85th percentiles, enrichment analyses were performed for Gene Ontology (GO) terms *(*ASHBURNER et al., 2000; THE GENE ONTOLOGY CONSORTIUM, 2023*)* and KEGG pathways (KANEHISA; GOTO, 2000). For the analysis, the "*org.Hs.eg.db*" database for *Homo sapiens* was used, accessed through the clusterProfiler package in R. The Fisher's exact test, with multiple testing correction using the Benjamini-Hochberg method (BENJAMINI; HOCHBERG, 1995), was applied to determine statistically significant enrichment of specific terms (pvalueCutoff = 0.01 e qvalueCutoff = 0.05).

2.2.3 Immune Pathways

To investigate evolutionary patterns in immune-related gene sets, two complementary approaches were applied:

1. Overall divergence of immune pathways: an exploratory analysis to estimate the typical divergence of each pathway across all 87 mammalian species. Using the median genetic distances of each gene (as described in section 2.2.1), we computed the median distance for each immune pathway in the KEGG and Reactome databases based on its constituent genes.

2. Species-specific divergence: The second strategy focused on identifying species-specific divergence signals. For this, genes were ranked by their genetic distance to the human sequence in each species, and a Gene Set Enrichment Analysis (GSEA) was performed independently for each species to identify pathways enriched among the most divergent genes.

2.2.3.1 Overall Divergence of Immune Pathways

This analysis focused on immune pathways to determine the evolutionary patterns and central tendencies of evolutionary distances among specific functional gene sets. Given the specific interest in immune-related genes, an analysis of immune pathways was carried out as follows, using two biological pathway databases: Reactome and KEGG.

1. Selection of Immune-related Pathways: From the KEGG database, accessed via KEGGREST package in R, all pathways under the "Immune System" category were selected, along with additional relevant pathways such as "Cell Adhesion Molecules," "JAK-STAT Signaling Pathway," "Cytokine-Cytokine Receptor Interaction," "Viral Protein Interaction with Cytokine and Cytokine Receptor," "NF-kappa B Signaling Pathway," "TGF-beta Signaling Pathway," and "TNF Signaling Pathway", resulting in 28 pathways. Utilizing GMT files from GSEA (SUBRAMANIAN et al., 2005) and referencing Reactome's pathway data (JASSAL et al., 2020), the focus was exclusively on terms under the "Immune System" category. This resulted in the inclusion of 159 pathways.

2. Calculation of Distance Estimates per Pathway: For each immunological pathway, the genes present in both the pathway and the distance dataset were identified and the median of the genetic distances of these genes was calculated, thus obtaining a representative measure of the average evolutionary divergence of the genes in each pathway.

3. Coverage: The coverage of each pathway was calculated, defined as the proportion of genes in the pathway that had genetic distance data available.

4. Analysis of the dispersion of the distances: The interquartile range (IQR) was calculated to measure the dispersion of the data around the median, providing information on the variability within each immune pathway.

5. Confidence interval of the median: The 95% confidence interval for the median of the genetic distances of each pathway was calculated using the bootstrap method with 1,000 resamplings. This procedure allows estimating the variability of the median, considering the sample size (number of genes in the pathway), and assessing the precision of the estimates obtained. A narrow confidence interval indicates a more precise estimate of the median, while a wider interval suggests greater variability in the data or a smaller number of genes contributing to the median.

Only pathways with at least five genes were included in the analysis. To visualize the differences in the estimates of evolutionary distances within the selected pathways in KEGG genesets, the Pathview package (LUO; BROUWER, 2013) in R was used. Genes within each pathway were color-coded based on their evolutionary distances, highlighting those with significant evolutionary divergence. These results allow the identification of the most conserved or divergent immune pathways, enhancing the understanding of evolutionary patterns in the mammalian immune system and highlighting the genes contributing most to these differences.

2.2.3.2 Species-Specific Analyses

GSEA was performed separately for each of the 87 species. For each species, orthologous genes were ranked by their genetic distance from the *Homo sapiens* sequence, with the most divergent genes placed at the top of the list. This approach adapts the GSEA framework, usually applied to expression data, to evolutionary divergence by using genetic distance as the ranking metric, allowing the identification of biological pathways enriched with genes showing higher evolutionary divergence.

Enrichment analysis was performed on each ranked gene list using the fgseaMultilevel function from the fgsea package in R (SERGUSHICHEV, 2016). Gene sets were obtained from the complete Reactome pathway database version 2024 (JASSAL et al., 2020). To identify pathways enriched with highly divergent genes, the scoreType parameter was set to "pos". Pathways containing fewer than 15 or more than 2,000 genes were excluded. Enrichment was considered significant when the adjusted p-value using the Benjamini-Hochberg method (BENJAMINI; HOCHBERG, 1995) was ≤ 0.05. To ensure a genome-wide analysis, it was used the full set of Reactome pathways, totaling 1097 pathways.

2.2.4 Analysis of Cytokine Receptors and Ligands

Given the fundamental role cytokines play in the immune system, we conducted a focused analysis of cytokine receptors and their corresponding ligands to examine patterns of

evolutionary similarity and divergence. The list of cytokine and its receptors was obtained from the KEGG database (KANEHISA; GOTO, 2000) (entry hsa4060: Cytokine-Cytokine receptor interaction). Genetic distances estimates, were extracted for each gene and compared across the dataset. The receptors and ligands with the highest median evolutionary distances were identified, indicating immune system components that may be under distinct selective pressures or involved in specific adaptive responses.

2.3 Analysis of Non-synonymous to Synonymous Substitution Ratios (dN/dS)

A site-specific selection analysis was done on genes showing high divergence to test whether they were accumulating an excess of nonsynonymous substitutions relative to synonymous ones, which could indicate adaptive evolution. In addition to testing for evidence of positive selection, the analysis sought to identify specific sites evolving under positive selective pressure and to determine which genes harbored the greatest number of such sites.

A set of 15 mammalian species was selected, prioritizing high-quality genome assemblies and broad phylogenetic coverage. The panel included primates (*Homo sapiens*, *Callithrix jacchus, Macaca mulatta, Pan troglodytes*), artiodactyls (*Bos taurus, Sus scrofa, Capra hircus*), a perissodactyl (*Equus caballus*), carnivores (*Canis lupus familiaris, Felis catus*), a lagomorph (*Oryctolagus cuniculus*), rodents (*Mesocricetus auratus, Mus musculus, Rattus norvegicus*), and a chiropteran (*Rhinolophus ferrumequinum*).

The analysis was done with codeml from the PAML package (YANG, 2007). Codeml applies codon-based substitution models to estimate the ratio of nonsynonymous to synonymous substitution rates (dN/dS or ω), with $\omega > 1$ indicating positive selection (YANG; NIELSEN, 2002; YANG, 2007; ÁLVAREZ-CARRETERO; KAPLI; YANG, 2023). As input, codeml requires a codon alignment and a phylogenetic tree. The analysis returns estimates of the dN/dS ratio and sites potentially under positive selection. A bioinformatic workflow was developed for this analysis and automated with the Snakemake workflow manager (KÖSTER; RAHM, 2012). The pipeline integrated all steps from coding sequence (CDS) collection to PAML analyses. All scripts are publicly available in a GitHub repository (https://github.com/MaiaraMoro/dnds-pipeline). The main steps are described below.

2.3.1 Gene Subset Definition

The candidate gene set was defined using the following approaches:

1. **Pathway-driven set:** In the GSEA results (section 2.2.3.2), the Reactome pathway *Interferon Signaling* was among the highest normalized enrichment scores (NES). Given its central role in antiviral defense and the interspecific variation in interferon pathways and antiviral outcomes (GONZALEZ et al., 2025; ZHOU et al., 2016), this pathway was selected for targeted analysis. All genes annotated in Reactome as belonging to *Interferon Signaling* were selected; from these, only one-to-one orthologs with reliable sequence data and genetic distance estimates (as described in section 2.1) were retained. This resulted in a final set of 190 genes included in the codeml analyses.

2. **Cross-species high-divergence set:** For each of the 14 species (*Callithrix jacchus, Macaca mulatta, Pan troglodytes*, *Bos taurus, Sus scrofa, Capra hircus*, *Equus caballus*, *Canis lupus familiaris, Felis catus*, *Oryctolagus cuniculus*, *Mesocricetus auratus, Mus musculus, Rattus norvegicus*, *Rhinolophus ferrumequinum*), genes were ranked by their genetic distance from *Homo sapiens* and those in the 95th percentile were selected. Genes that appeared in at least half of the species (≥ 7 of 14) were retained, totalling 354 genes. Functional annotations from GO (BOTSTEIN et al., 2011) and GeneCards (SAFRAN et al., 2010) were used to retain only immune-related genes. This filtering yielded 94 candidates, which were then tested for positive selection with codeml.

For comparison, two additional gene sets were included alongside the interferon and high-divergence immune groups, to contrast patterns of positive selection in highly conserved genes and in a neutral random sample:

3. **Cross-species conserved set:** Using the same approach, for each of the 14 species genes in the 5th percentile of genetic distances to *Homo sapiens* were identified. Genes consistently falling in this lower tail in at least half of the species (≥ 7 of 14) were retained as conserved candidates. From this pool, 92 genes were randomly selected to generate a set of highly conserved genes for comparative analyses.

4. **Random set:** As a neutral reference, 87 genes were randomly sampled from the complete pool of 16,210 one-to-one orthologs.

Together, these four gene sets (190 interferon, 94 high-divergence immune, 92 conserved, and 87 random) were analyzed with codeml and used for comparative enrichment analyses of positive selection. Statistical comparisons between sets were conducted using Fisher's exact test. Because not all 14 species had one-to-one orthologs for every gene, the number of species' sequences analyzed varied among genes. The number of species and the genes analyzed is provided in Supplementary Table S1.

2.3.2 Orthologous Coding Sequences (CDS) Collection and Codon Alignment

For each human gene of interest, the corresponding one-to-one orthologous coding sequences (CDS) from 14 other mammalian species were retrieved from Ensembl (release 113). CDS sequences that failed to produce a valid translation (e.g., due to premature stop codons) were discarded. Protein sequences were aligned with PRANK (LÖYTYNOJA; GOLDMAN, 2005) using default parameters. The resulting protein alignment was then used as a guide to generate the codon alignment. Specifically, the original unaligned CDS were mapped onto the protein alignment using the Pal2Nal.pl script (SUYAMA; TORRENTINO; HIRASAWA, 2006). This step ensures that the alignment reflects evolutionary changes at the codon level.

2.3.3 Phylogenetic Inference

A maximum likelihood phylogenetic tree was inferred from each codon alignment using IQ-TREE 2 (MINH et al., 2020). The best-fitting substitution model was selected with ModelFinder Plus (-m MFP), which compares a set of evolutionary models and chooses the one that best explains the data according to the Bayesian Information Criterion (BIC). The resulting gene trees, optimized with the chosen models, were then used in downstream analyses of selection.

2.3.4 Detection of Site-Specific Positive Selection

The detection of sites under positive selection was done with codeml by fitting and comparing alternative evolutionary models. It was evaluated the models M0 (one ω for all sites), M1a (neutral, allowing only purifying selection or neutrality), M2a (allowing an additional class of sites with $\omega > 1$), M7 (beta distribution of ω between 0 and 1), and M8 (as M7 but with an extra category of sites with $\omega > 1$) (ÁLVAREZ-CARRETERO et al., 2023). Model comparisons between M1a (nearly neutral) vs M2a (positive selection) and M7 (beta) vs M8 (beta&ω*)* were carried out using likelihood ratio tests (LRT), where twice the log-likelihood difference was compared to a χ^2 distribution with 2 degrees of freedom. Resulting p-values were adjusted for multiple testing using the Benjamini-Hochberg method to control the false discovery rate (FDR). A gene was considered to be under positive selection when the M7 vs. M8 comparison was significant ($q \leq 0.05$). For these genes, individual sites under positive selection were identified with the Bayes Empirical Bayes (BEB) approach under the M8 model, sites with posterior probability ≥ 0.95 were considered as having evidence of positive selection.

2.3.5 Detection of Branch-Specific Positive Selection

To complement the site-level analysis of positive selection, we performed branch-level tests on genes with the highest number of positively selected sites. This aimed to investigate whether selection was concentrated in specific species within the phylogenetic tree. For this, we applied the Adaptive Branch-Site Random Effects Likelihood (aBSREL) method implemented in HyPhy (SMITH et al., 2015), which detects episodic positive selection acting on individual branches of a phylogeny.

Codon alignments were generated as described above and provided in PHYLIP format. Maximum-likelihood phylogenetic trees inferred with IQ-TREE (section 2.3.3) were used as fixed topologies for each gene. The analysis was restricted to terminal branches

(*--branches Leaves*) to test for lineage-specific selection. Branches with significant adjusted p-values ($q \leq 0.05$) were considered to have evidence of episodic diversifying selection.

2.4 PPI Network analysis

To integrate functional relationships of the genes with the evolutionary patterns identified in this study, a protein-protein interaction (PPI) network was constructed for the 190 genes of the reactome interferon signaling pathway (R-HSA-913531). The network was built using data from the STRING interactome database (SZKLARCZYK et al., 2021), imported via NetworkAnalyst (https://www.networkanalyst.ca/). Only direct (zero-order) interactions with a high confidence score (> 0.7) were included. The resulting network was visualized and analyzed in Cytoscape (SHANNON et al., 2003). A topological analysis was done using the NetworkAnalyzer plugin to calculate connectivity metrics, with the degree of each node used to identify highly connected proteins (hubs). Densely connected modules or communities were identified using the clusterMaker2 with the GLay Community Clustering algorithm (SU et al., 2010).

Finally, findings of evolutionary data were mapped onto the network as visual attributes. The color of each node was mapped to its median genetic distance (as calculated in section 2.1.3). Also, genes with evidence of positive selection and at least one site inferred by BEB (≥0.95) were displayed in diamonds shape, and the size of each node was made proportional to its degree, emphasizing the network hubs. This visualization allows for the simultaneous assessment of a gene's functional connectivity and the overall level of evolutionary divergence between different species to *Homo sapiens*.

2.5 Species-level regression analysis

To investigate if some species accumulate substitutions more strongly than others as genetic divergence increases, we regressed relative divergence (Δ) on overall gene divergence for each species and compared the resulting slopes (β). The regression slope (β) captures how strongly each species' divergence from humans increases as genes become more divergent,

reflecting lineage-specific patterns of substitution accumulation. This allows comparison of divergence patterns across species and taxonomic orders.

Overall divergence for each gene was defined as the median genetic distance to *Homo sapiens* across species (as described in section 2.2.1). We standardized it across genes using z-scores (subtracting the mean across genes and dividing by the standard deviation).

Relative divergence was defined, for gene *g* and species *s*, as:

$$\triangle_{g,s} = d_g(s, H) - d_g(C, H)$$

d = evolutionary distance
g = gene
S = species
H = *Homo sapiens*
C = Chinpanzee

This metric quantifies how much more (or less) divergent a species is from humans compared to the chimpanzee. We used *P. troglodytes* as the reference due to its close evolutionary relationship to humans, providing a stable baseline.

The analysis was performed on two gene sets: the complete set of 15,430 genes with available data for *Pan troglodytes* and a subset of 2,205 genes annotated to the "immune system process" category of the GO (GO:0002376). For each of the 86 mammalian species, we computed two regression slopes (β): based on the full gene set (β all) and only on immune-related genes (β immune). To test if immune genes to show higher slopes than the complete set of genes, we compared β immune and β all values within each species using a one-sided paired Wilcoxon signed-rank test (H1: β immune > β all).

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